

Enumerated-reading evidence-chain migration and re-validation record (migration batch 2)

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This note is the permanent record of migration batch 2 (plan phase M1): the enumerated-reading / estimator evidence chain of claim **B1-RH-ENUM** (Reading-H selection within enumerated condensate ensembles, T5 CLOSED@ESTIMATOR-GRADE) was copied verbatim from the frozen legacy repository into [archive/legacy/](#) and re-validated by full re-execution. With this batch **B1-RH-ENUM** is *migration-clean* (no unresolved **legacy:** pointers). The same chain grounds the justification of hypothesis H-LAYER carried by **B2-PROPA-HLAYER**. This note makes no new mathematical claim and performs no tier action.

2. What was migrated

32 files: 14 notes (Math427 v1.0/v1.1, Math428 v1.0/v1.1, Math429 v1.0/v1.1, Math430 v1.0, Math431 v1.0, Math432 v1.0/v1.1, Math434 §15.5 audit + AddA T5-promotion record, Math436 v1.0/v1.1), 8 verification scripts, and 10 archived artefacts (8 result JSONs + the Math434/Math436 checkpoint `state.json` files, kept as provenance). Layout: [archive/legacy/{notes/<Tag>/, scripts/, artefacts/<Tag>/}](#); original paths per file in [archive/MIGRATION-LEDGER.md](#).

3. Numerical verification

Fresh environment: python 3.12, numpy 2.2.6. Every script re-executed from scratch (no legacy checkpoint state used).

Script	Asserts	Verdict line
Math427_g1prime_diagonal_isotropy.py	5/5	diagonal-isotropy grid minimum nonnegative
Math428_gidoubleprime_bloch_logdet.py	21/21	Bloch log-det race PASS, continuum-anchored, both bands positive
Math429_g1pp1prime_inhomogeneous_wick.py	19/19	anchored exact-Wick M -scan stays nonnegative
Math430_g1pp2_dense_surface_convergence.py	11/11	dense surface positive; convergence spots positive
Math431_g1pp3_lam_hex_fcc.py	15/15	LAM / HEX / FCC / FCC-exact-Wick all PASS
Math432_g3prime_multishell_ensemble.py	25/25	two-shell surface positive everywhere
Math434_lam_exact_wick_bracket.py	22/22	LAM exact-Wick anchored positive at every grid point
Math436_hex_exact_wick_bracket.py	49/49	HEX exact-Wick anchored positive at every grid point

Total: **167/167 asserts PASS** (5+21+19+11+15+25+22+49). All eight regenerated JSONs are identical to the archived legacy artefacts within relative tolerance 10^{-9} (absolute 10^{-12}): *zero diffs, zero stale-artefact findings* — in contrast to batch 1’s finding F-1. Fresh artefacts: [runs/B1-RH-ENUM/260605-migration-revalidation/](#) (8 JSONs + [summary.json](#)).

Quantitative sanity check (mandatory): the assert total was cross-checked against the per-script verdict lines (167); the comparison script reported 0 numeric and 0 string differences; runtimes $O(1\text{--}40\text{s})$ per script, consistent with grid sizes.

4. Findings

None. No stale artefacts, no convention drift (all notes are themselves the corrected-convention $r_{\text{braz}} = K(q_0) = \mu^2$ lineage), no assert failures.

5. Hypothesis-chain linkage

H-LAYER (claims/GATES.md) justifies the isotropic Gaussian–Hartree layer by (i) the diagonal-isotropy infimum (Math427) and (ii) beyond-diagonal refinements EXECUTED for the five enumerated readings (Math428–432, Math434, Math436). Both legs are now migrated and re-validated; the unexecuted non-enumerated residual remains exactly gate STEP-5B, unchanged by this batch.

6. Devil’s-advocate (three objections)

- (α) “*Same-ecosystem re-execution is not independent verification.*” VALID with mitigation: the asserts are the original authors’ self-tests; reproducibility, not correctness, is certified here. Independent correctness evidence is the migrated Math434 §15.5 independent-audit note; CI re-execution on the GitHub runner adds an environment-independent check.
- (β) “*Math434/Math436 completed in a single budget window, so the checkpoint-resume path went unexercised in this re-validation.*” VALID: recorded as a limitation. The resume path is evidenced by the archived [state.json](#) provenance files; it is falsifiable at any time by running under a tighter budget and observing a correct resume.

(γ) “Enumerated coverage does not establish admissible-class exhaustiveness.” DISMISSED as an objection to this record: the record claims migration and reproducibility only. Exhaustiveness is exactly the open gate STEP-5B, explicitly out of scope; B1-RH-ENUM remains T5 CLOSED@ESTIMATOR-GRADE.

7. Result footer

Result ID:	B1-RH-ENUM (supporting: H-LAYER of B2-PROPA-HLAYER)
Precise statement:	Migration batch-2 evidence chain migrated verbatim and re-validated by full re-execution (167/167).
Scope:	Reproducibility of the migrated chain only; no new mathematical claim; no tier action.
Dependencies:	archive/legacy/ batch-2 file set (ledger 2026-06-05).
Evidence grade:	EXECUTED
Reproduction command:	cd archive/legacy/scripts && python Math427_... && ... && Math436_... (8-script chain; full command on the B1-RH-ENUM claim card)
Expected output:	5/5, 21/21, 19/19, 11/11, 15/15, 25/25, 22/22, 49/49 asserts PASS; JSONs match archive at 1e-9; Math434/436: re-run on [CHECKPOINT] until VERDICT
Falsification gate:	Any assert failure or artefact mismatch beyond tolerance on re-execution.
Tier before / after:	B1 T5 / T5 (no change)
No-overclaim statement:	Reproducibility, not correctness; enumerated ensembles only -- Reading-H stays T5; STEP-5B open.
Next required action:	STEP-5B beyond-layer class-wide bound.